Reproducibility: an open-source tool for computational hypothesis testing in natural language

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Summary

We present *Reproducibility*, an open-source software system for quantitative evaluation of scientific hypotheses formulated in natural language. The system converts scientific statements written in Portuguese or English into executable mathematical functions, assesses their internal consistency through Monte Carlo simulation, and measures their predictive power against real empirical data from open sources (World Bank, NASA POWER, IBGE) or user-supplied CSV files. We propose a *composite reproducibility score* — combining simulated consistency and empirical fit — and demonstrate its validity against five cases with expected behaviour established by the literature: the CO_2–global temperature relationship (composite score = 49.1; expected ≥ 40), the minimum wage effect on employment from Card & Krueger (composite score = 0.3; expected ≤ 20), the replication rate of psychological studies from the Open Science Collaboration (composite score = 20.8; expected 10–30), the Preston Curve relating GDP per capita to life expectancy (composite score = 56.3; expected ≥ 40), and the Kuznets Environmental Curve relating GDP per capita to CO_2 emissions per capita (composite score = 48.3; expected ≥ 40). The tool requires no proprietary software, operating with open-source language models (Llama 3, Mistral) via Ollama or the free Groq API, or with an entirely offline rule-based parser.

1 Statement of Need

The reproducibility crisis in science is a documented, large-scale problem. Ioannidis (2005) showed through probabilistic analysis that the majority of published findings in biomedical research may be false, considering the combined effects of insufficient sample sizes, analytical flexibility, and publication bias. This prediction was empirically confirmed by Open Science Collaboration (2015), who conducted replications of 100 psychology studies and found that only 36% reproduced statistically significant results, with replication effects approximately half the original magnitude.

There is an important conceptual distinction between *reproducibility* — obtaining the same results with the same data — and *replicability* — obtaining equivalent results

with new independent data (Goodman, Fanelli, and Ioannidis 2016). Crucially, both concepts operate *post hoc*: they require data to already exist. Our tool operates in a third, *a priori* domain: given that a researcher formulates a hypothesis, it provides a quantitative estimate of whether the mathematical structure of that hypothesis is compatible with stable, replicable results — *before any data collection occurs*. This distinction is the primary differentiator from all existing tools.

Existing tools for reproducibility support focus primarily on *post hoc* verification: the `repro` R package (Peikert and Brandmaier 2021) allows previously performed analyses to be reproduced with the same data and code, and platforms such as OSF (Center for Open Science 2013) facilitate data and code sharing after publication. `ReproducibiliTeachR` is an R package designed for teaching reproducibility concepts rather than evaluating research hypotheses. We have not identified in the literature a tool that allows a researcher to write a hypothesis in natural language and receive, within seconds, a quantitative estimate of its expected reproducibility against open data before the study is conducted. This gap motivates the present contribution.

2 Theoretical Background

2.1 Simulated reproducibility score

The *simulated reproducibility score* S_{sim} is defined as:

$$S_{\text{sim}} = 100 \cdot \exp\left(-\frac{CV}{10}\right) \tag{1}$$

where CV is the Coefficient of Variation (standard deviation / mean $\times$ 100%) of the outputs from n independent Monte Carlo simulation trials. Each trial samples values of the hypothesis variables from their plausible ranges and computes the output of the corresponding mathematical function.

To illustrate the extremes: a deterministic physical law such as Newton's gravitational law has $CV \approx 0\%$ and $S_{\text{sim}} = 100$, while a hypothesis dominated by structural noise (e.g., a social outcome driven primarily by factors outside the model) may have $CV > 40\%$ and $S_{\text{sim}} < 1$.

The function has the following desirable mathematical properties:

1. $S_{\text{sim}} \in [0, 100]$: normalised score, comparable across hypotheses
2. $S_{\text{sim}}(CV = 0) = 100$: zero variation implies perfect reproducibility
3. $S_{\text{sim}}(CV \to \infty) \to 0$: unbounded variation implies non-reproducibility
4. Strict monotonicity: increasing CV implies decreasing S_{sim}

2.2 Composite score

A fundamental limitation of S_{sim} in isolation is that hypotheses with simple deterministic functions — including trivially vacuous ones such as the constant function $y = 5$ — obtain $S_{\text{sim}} = 100$ regardless of their predictive power over real data. To also capture empirical fit, we define the *composite score* as the primary reporting metric:

$$S_{\text{comp}} = S_{\text{sim}} \cdot \max(0,\ R^2) \tag{2}$$

where R^2 is the coefficient of determination obtained by testing the hypothesis against an empirical dataset. This formula penalises hypotheses that, while mathematically consistent, do not explain the observed variation in data. When $R^2 \leq 0$ — meaning the hypothesis performs worse than simply predicting the mean — the composite score is zero. The composite score is automatically computed by `run_pipeline()` and is the recommended metric for all reporting. S_{comp} is a scale on $[0, 100]$ and is **not** a probability; it should not be interpreted as or compared directly with the Akaike weights w_i, which are probabilities summing to one across the competing hypotheses.

2.3 Hypothesis comparison via Akaike criterion

To rank competing hypotheses explaining the same phenomenon, we use the Akaike information criterion (Burnham and Anderson 2002):

$$AIC = 2k - 2\ln(\hat{L}) \tag{3}$$

AIC was selected over BIC and adjusted-R^2 because the hypotheses tested are typically non-nested — none is a special case of another — a condition for which AIC was specifically designed (Burnham and Anderson 2002). The implementation omits additive constants from the Gaussian log-likelihood ($-\frac{n}{2}\ln(2\pi)$ and $-\frac{n}{2}$) that cancel in all Δ_i computations and therefore do not affect Akaike weights or rankings. Absolute AIC values are consequently not comparable across datasets of different size n; only Δ_i and w_i should be interpreted. The normalised Akaike weight w_i represents the relative probability that each hypothesis is the best approximation to reality given the tested model set:

$$w_i = \frac{\exp(-\Delta_i/2)}{\sum_j \exp(-\Delta_j/2)} \tag{4}$$

When $\Delta_i < 2$, competing hypotheses are considered statistically indistinguishable and both are reported as plausible. This behaviour was observed empirically in Case 5 of the validation (Section 4), where linear and logarithmic hypotheses for CO_2–GDP yielded $\Delta\text{AIC} = 1.9$, correctly reflecting the genuine functional ambiguity documented in that literature.

3 Software Architecture

The system is organised into two main files. **`reproducibility_core.py`** is a pure logic module with no UI dependencies, importable in any Python environment as a standard library — enabling integration into automated pipelines and research scripts independently of the web interface. It implements eight subcomponents: data structures (`Variable`, `Hypothesis`, result types); `OpenTranslator` for three-layer NLP translation; `ReproducibilityEngine` for Monte Carlo simulation; `EmpiricalTester` for fit to real data with R^2, RMSE, AIC, and BIC; `HypothesisComparator` for Akaike

criterion ranking; `TemporalAnalyser` for drift detection; `DataFetcher` for unauthenticated open API access; and `generate_dashboard` for matplotlib visualisation. `app.py` is a Streamlit web interface with three tabs: hypothesis definition, data selection, and execution with result visualisation.

3.1 Module 1: Hypothesis translation (Open Science)

The `OpenTranslator` operates in three layers with automatic fallback. **Layer 0 — RuleParser** is always available, requires zero dependencies, and works offline. It uses a regex-based system with a dictionary of 60+ scientific variables in Portuguese (PT-BR) and English, recognising eight mathematical relationship types (linear, inverse, power, exponential, logarithmic, constant, quadratic, cubic). The bilingual coverage makes the tool directly accessible to researchers in Brazil and other Lusophone countries without requiring an LLM. **Layer 1 — Groq API** accesses Llama 3.3 70B (Meta AI 2024) and Mixtral 8×7B (Mistral AI 2023) via a free API without a credit card. Both models are open-source (Meta Llama licence and Apache 2.0, respectively), consistent with the Open Science principles of the tool. **Layer 2 — Ollama** runs open-source models locally on the researcher's own machine without sending data to external servers, recommended for sensitive research data.

3.2 Open data sources

Source	Type	Authentication	Coverage
World Bank API	Global indicators	None	200+ countries, 1960–present
NASA POWER	Climate data	None	Any coordinate, 1981–present
IBGE SIDRA	Brazilian data	None	Brazil, historical series
CSV / Excel	User data	—	Any domain

Table 1: Open data sources integrated in the `DataFetcher` module.

4 Validation

The validation is structured in two phases. **Phase 1** (Table 2) uses cases with answers known *a priori* from mathematics or physical theory, verifying that the instrument is correctly calibrated at the extremes of the scale (score = 100 for deterministic laws; score ≈ 0 for pure noise). **Phase 2** (Table 3) uses published empirical datasets with expected score ranges established by the literature, verifying that the instrument discriminates correctly across the full spectrum from controversial to robust relationships.

4.1 Cases with known *a priori* answers

The third case implements a simplified Mincer earnings equation (Mincer 1974; Becker 1964), modelling income as a linear function of education and experience with a large stochastic residual ($\sigma = 3000$ monetary units) representing structural factors outside the model (discrimination, inheritance, macroeconomic shocks). The dominant noise term produces a high CV and a near-zero score, confirming that the

Hypothesis	Type	S_{sim}	CV (%)	Expected
Newton's Law of Universal Gravitation	Physical law	100.0	0.000	≥ 85
Constant function ($y = 5$)	Constant	100.0	0.000	≥ 95
Mincer earnings equation + high noise (Mincer 1974)	High-noise social	0.8	48.8	≤ 10
Hypothesis with noise $\sigma = 1000$	No structure	0.0	2129	≤ 10

Table 2: Cases with known *a priori* answers. Parameters: $n = 100$ trials $\times$ 500 samples, seed = 42. These higher-precision parameters are used in the `TestScoreValidation` test class to serve as calibration anchors. Table 3 results were produced with $n = 80$ trials $\times$ 300 samples.

instrument correctly penalises hypotheses whose explanatory variables account for only a small fraction of outcome variance.

4.2 Validation against published article datasets

Dataset	Source	Best hypothesis	S_{sim}	R^2	S_{comp}	Expected	Status
$CO_2 \rightarrow$ Global temp.	IPCC AR6 / NASA GISS	Linear	69.0	0.711	**49.1**	≥ 40	✓
Min. wage $\rightarrow$ Employment	Card and Krueger (1994)	Positive effect	99.6	0.003	**0.3**	≤ 20	✓
Original $\rightarrow$ Replication	Open Science Collaboration (2015)	50% attenuation	73.1	0.285	**20.8**	10–30	✓
GDP $\rightarrow$ Life expectancy	WB / Preston (1975)	Logarithmic	96.4	0.584	**56.3**	≥ 40	✓
GDP $\rightarrow$ CO_2 emissions	WB / Grossman and Krueger (1991)	Linear	78.2	0.617	**48.3**	≥ 40	✓

Table 3: Validation against published article datasets. Cases 1–3: datasets reconstructed from published statistics. Cases 4–5: World Bank Open Data averages (2010–2022 and 2015–2020 respectively). WB = World Bank.

The CO_2–temperature case (score = 49.1) confirms the well-documented robustness of the relationship (IPCC 2021). The Card & Krueger case (score = 0.3) reflects that the minimum wage effect on employment is essentially negligible relative to the natural variance in restaurant-level employment data — the best hypothesis (positive effect) explains only 0.3% of the outcome variance ($R^2 = 0.003$), consistent with decades of controversy about whether the effect is real (Card and Krueger 1994; Neumark and Wascher 2000). The OSC 2015 case (score = 20.8) is consistent with the finding that replication effects are approximately half the original magnitude (Open Science Collaboration 2015).

The Preston Curve case (score = 56.3) confirms the well-established relationship between national income and population health (Preston 1975), with the logarithmic form correctly identified as the best functional form — consistent with the diminishing marginal returns to health outcomes from income at higher development levels. The CO_2–GDP case (score = 48.3) is consistent with the empirical literature on the Kuznets Environmental Curve (Grossman and Krueger 1991); the near-equivalent fit of linear and logarithmic hypotheses (ΔAIC = 1.9) correctly reflects the genuine functional ambiguity documented in that literature for the income range covered by the dataset.

A key methodological finding is that the separation between S_{sim} and S_{comp} dis-

tinguishes *internal consistency* from *empirical validity* — two dimensions that the reproducibility literature treats separately (Goodman, Fanelli, and Ioannidis 2016). A hypothesis can be highly reproducible in simulation yet fail empirically, as illustrated by the constant-function baseline in each case.

Note on negative R^2: in Cases 1 and 3, some competing hypotheses yield $R^2 < 0$, meaning the model performs worse than simply predicting the unconditional mean. In Case 1, the Arrhenius logarithmic parametrisation, while physically motivated, is not calibrated to the 1900–2022 range where the linear approximation holds. In all such cases $S_{\text{comp}} = 0$ by the $\max(0, R^2)$ term.

5 Comparison with Related Tools

Feature	Reproducibility	repro (R)	ReproducibiliTeachR	OSF
Natural language hypotheses	✓	×	×	×
Quantitative reproducibility score	✓	×	×	×
Hypothesis comparison (AIC)	✓	×	×	×
Temporal drift analysis	✓	×	×	×
Integrated open API data	✓	×	×	✓
Reproduction of published analyses	×	✓	×	✓
Web interface	✓	×	✓	✓
No proprietary software	✓	✓	✓	✓
No cost	✓	✓	✓	✓

Table 4: Comparison with related tools. ✓ = feature present; × = feature absent.

The `repro` package and OSF excel at verifying whether an *already performed* analysis can be reproduced with the same data — a post hoc use case for which this tool is not designed. The two approaches are complementary: `repro` and OSF verify computational reproducibility after a study is complete; this tool estimates replicability potential before the study begins. Note that only this tool operates a priori and provides a quantitative score.

6 Usage Example

Listing 1: Minimal working example: hypothesis translation, World Bank data retrieval, and full pipeline execution.

```
from reproducibility_core import OpenTranslator, DataFetcher
    , run_pipeline

# 1. Hypothesis in natural language (English or Portuguese)
```

```python
translator = OpenTranslator()
hypothesis = translator.translate(
    "Life expectancy increases with GDP per capita"
)

# Always inspect variable symbols before constructing the
   mapping.
# The RuleParser may detect multiple variables; all must be
   mapped.
for v in hypothesis.variables:
    print(f"  symbol '{v.name}' -> {v.description}")
# Example output:
#   symbol 'ev' -> life expectancy (anos)
#   symbol 'gdp' -> gdp (USD)

# 2. Real data from World Bank (no API key required)
df = DataFetcher().fetch_worldbank_multiple_countries(
    indicators = ["life_expectancy", "gdp_per_capita"],
    countries  = ["BR", "US", "DE", "CN", "IN", "ZA"],
)

# 3. Full pipeline     map every detected symbol to a
   DataFrame column
results = run_pipeline(
    hypotheses    = [hypothesis],
    df            = df,
    target_column = "life_expectancy",
    mappings      = [{"ev": "life_expectancy", "gdp": "
       gdp_per_capita"}],
)
print(f"R2              = {results['empirical'][0].r_squared
   :.3f}")
print(f"Sim. score      = {results['simulation'][0].
   reproducibility_score:.1f}/100")
print(f"Composite score= {results['empirical'][0].
   composite_score:.1f}/100")
```

The `mappings` parameter bridges hypothesis variable symbols (assigned automatically by the translator) to the actual column names in the DataFrame. Always inspect `hypothesis.variables` before defining the mapping to confirm which symbols were detected — the RuleParser may identify multiple independent variables, all of which must appear as keys in the mapping dictionary. Mapping only a subset of detected variables will cause a `KeyError` at runtime.

7 Automated Tests

The test suite in `tests/test_core.py` contains 52 tests organised in eight classes covering all modules. The `TestScoreValidation` class is the scientific core of the suite: its six tests verify not only computational correctness but the scientific validity of the instrument — whether it correctly discriminates between the cases with known

answers in Table 1 (physical laws → score ≥ 85; high-noise social hypotheses → score ≤ 10; strictly monotonic decrease with added noise).

Run with: `pytest tests/ -v`

8 Limitations

The **RuleParser** (Layer 0) operates in a closed domain: variables absent from the internal dictionary are mapped to generic symbols. For specialised domains (biochemistry, astrophysics, clinical pharmacology), use of the Groq or Ollama layers is recommended. Additionally, when the RuleParser detects multiple variables in a hypothesis, all are treated as additive independent predictors; the user must verify the mapping between hypothesis variable symbols and dataset columns before running the pipeline (inspecting `hypothesis.variables` is recommended to avoid mapping errors).

The system internally computes a secondary diagnostic metric, `empirical_score`, combining R^2 (weight 0.70) and a residual-normality indicator (weight 0.30). These weights are heuristic and not derived from a formal criterion; the metric is exposed in the ranking table for diagnostic purposes only. The **composite score** (S_{comp}) is the theoretically grounded primary metric and should be used for all comparisons and reporting.

The **composite score** formula was validated on the ten cases reported in Tables 1 and 2, but has not been calibrated against a large corpus of published hypotheses; qualitative thresholds (≥ 40: robust relationship; 10–30: moderate; ≤ 10: weak or controversial) are provisional and may require domain-specific adjustment. Furthermore, S_{sim} measures *variability* but not *systematic bias*: a hypothesis may be highly internally consistent yet systematically wrong relative to empirical data; S_{comp} partially addresses this through R^2, but sensitivity to dataset quality and sample size is not quantified. No confidence interval is currently reported for either score; differences smaller than approximately 5 points should be interpreted with caution given Monte Carlo sampling variance. The Shapiro-Wilk normality test is applied to a maximum of 50 observations (trials or residuals); with larger samples, the test is known to detect trivially small deviations that have no practical significance, so this truncation is intentional (Burnham and Anderson 2002). Additionally, S_{sim} assumes a **uniform prior** over each variable's domain: all values in $[v_{\min}, v_{\max}]$ are sampled with equal probability. This simplification may over- or under-estimate the true CV when the variable's real-world distribution is non-uniform (e.g., concentrated near a mode). Users should set domain bounds to reflect the range of scientific interest rather than theoretical extremes.

Hypotheses with variable interactions (moderation, mediation) require the user to supply the functional form manually, as the current NLP layers do not parse interaction terms. This limits direct applicability to multivariate causal models common in psychology, medicine, and epidemiology.

When a fixed seed is used, the random number generator is reset to the same state at the start of each hypothesis evaluation. Consequently, the Monte Carlo trials of different hypotheses tested in the same session are **correlated**: they share the same random sequence. This ensures deterministic, reproducible scores for each hypothesis

individually, but means that score differences between hypotheses do not arise from statistically independent samples. For applications requiring inter-hypothesis independence (e.g., hypothesis ranking with formal uncertainty quantification), users should omit the seed or derive distinct seeds per hypothesis.

9 Future Work

- Calibration of the composite score against a corpus of hypotheses with known reproducibility (e.g., OSF studies with documented replication rates)
- Bootstrap confidence intervals for S_{sim} and S_{comp}
- Temporal windows defined by equal time intervals rather than equal row counts, to handle datasets with irregular temporal sampling
- `suggest_mapping()` helper function to automatically infer variable-to-column mappings from name similarity, reducing a common source of user error
- Integration with open repositories (Zenodo, Figshare, OSF) for automatic dataset retrieval by keyword
- Command-line interface (CLI) for use in scientific CI/CD pipelines
- Expanded variable dictionary for specialised domains (biochemistry, astrophysics, clinical pharmacology)

10 Dependencies

Package	**Min. version**	**Role**
numpy	1.24.0	Vectorised numerical operations
scipy	1.11.0	Statistics (Shapiro-Wilk, t-CI, regression)
pandas	2.0.0	Tabular data manipulation
matplotlib	3.7.0	Dashboard generation (Agg backend)
requests	2.31.0	Open APIs (World Bank, NASA POWER, IBGE)
streamlit	1.32.0	Web interface
openpyxl	3.1.0	Excel file support

Table 5: Python dependencies and minimum versions.

License

The software is released under the MIT License.

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